

# h\_da - IMSE - Embedded Signal Processing Systems (ESPS)

## General aspects

- The exam will be split in four parts with 100 points in total.
- This is a closed book exam. Students are only allowed to use a provided SystemVerilog help-sheet.
- In case you submit multiple solutions, you must clearly indicate which one should be used for evaluation. Otherwise, no solution will be considered and the question is marked with zero points.
- To get full credit for a problem you must show the details of your work. The maximum number of points per question is indicated in square brackets after each question.
- Exam duration: 90 minutes. Only in case you are well prepared, you will be able to answer all questions.
- Level of difficulty:

40 %  
Basic

40 %  
Intermediate

20 %  
Advanced

- Basic: You understood the respective lecture topics.
- Intermediate: You were able to solve the respective questions presented in class/lecture slides.
- Advanced: You are able to apply the key concepts in a new and unknown context.

## Direct Digital Frequency Synthesis – Part #1 (35%)

SystemVerilog, block diagrams, comprehension questions

- You should be able to explain the respective DDFS parameters in their respective role in the overall system.
- You should be able to explain possible resource-related optimization strategies.
- You should be able to use a given DDFS systems in an application specific context.
- You should be able to describe the effects of phase truncation on the spectral purity of the synthesized sine wave.
- You should be able to explain possible circuit extensions in order to improve the SFDR of the synthesized sine wave.

## Direct Digital Frequency Synthesis – Part #2 (15%)

SystemVerilog, block diagrams, comprehension questions

- You should be able to implement a dedicated DDFS configuration according to a given specification using SystemVerilog.

## Discrete-Time Systems - Analysis and Implementation – Part #1 (35 %)

SystemVerilog, block diagrams, comprehension questions

- Analysis of simple digital filters (this also includes the recursive oscillator structure discussed).
- Why do we use the Q1.15 fixed-point numbering format? What are the respective advantages?
- You should be able to implement a 'time base generation system' to generate the sampling rate for a given digital filter.
- You should be able to derive the difference equation/transfer function/ impulse response for a given system structure and to roughly sketch the corresponding frequency response.
- You should be able to design a certain system according to a given difference equation/transfer function.
- You should be able to apply simple optimization strategies in order to reduce the amount of computations involved.

## Discrete-Time Systems - Analysis and Implementation – Part #2 (15%)

SystemVerilog, block diagrams, comprehension questions

- You should be able to implement a dedicated discrete-time system according to a given specification using SystemVerilog.